

## Multiple Choice (10 marks)

1. PreP is an effective strategy for reducing the incidence of HIV in:
  - (a) Women
  - (b) Drug users
  - (c) Pregnant women
  - (d) a. and b. above
2. Laryngeal oedema is usually confined to the larynx above the vocal folds because
  - (a) there are numerous mucous glands in the vestibular folds.
  - (b) the mucosa covering the vocal folds is tightly attached to underlying tissues.
  - (c) fluid will drain rapidly into the thorax below the vocal folds.
  - (d) the mucosa above the vocal folds is more vascular than that below the vocal folds.
3. Deep nucleotide sequencing (NGS) is now cheap enough for general application. What will the technique mainly be useful for?
  - (a) Speedy vaccine development
  - (b) Identifying genetic variation
  - (c) Giving details on virus excretion in symptomless carriers
  - (d) Quickly identifying new viruses
4. All other things being equal, which of the following persons is more likely to show osteoporosis?
  - (a) An older Hispanic American woman
  - (b) An older African American woman
  - (c) An older Asian American woman
  - (d) An older Native American woman
5. Asp 235 Phe in a molecular report indicates that:
  - (a) asparagine has been replaced by phenylalanine.
  - (b) phenylalanine has been replaced by asparagine.
  - (c) aspartic acid has been replaced by phenylalanine.
  - (d) phenylalanine has been replaced by aspartic acid.

## True / False (10 marks)

*Answer True or False*

6. True or False: Proprioceptive nerve endings in synovial joints are located in the articular cartilage and synovial membrane.
7. True or False: A CVA in the left internal capsule can lead to spastic paralysis of the right leg due to disrupted motor neuron pathways.
8. True or False: The original rabies vaccine was made from air-dried spinal cords of infected rabbits, which served as a killed vaccine.
9. True or False: The risk of abnormality in the child of a mother with untreated phenylketonuria is almost 100%.
10. True or False: Around 20% of adolescent males have homosexual experiences during their teenage years.

## Long Form Response (20 marks)

*Answer each question with a clear and complete explanation.*

11. Discuss the advantages and limitations of using a quasi-experimental approach in health sciences research when comparing differences between two or more groups.
12. Discuss the replication process of picornaviruses as positive strand RNA viruses, focusing on the role of virion RNA as mRNA and its implications for viral protein synthesis.

*End of Paper*